

SymPy Bug Card

Bug 31: False Real Solutions for $\text{acos}(\cos(u)) = -1$

Task. Solve the real equation

$$\text{acos}(\cos u) = -1.$$

SymPy's answer. `solveset(acos(cos u) = -1, u, ℝ)` returns

$$\{2\pi n + 1 : n \in \mathbb{Z}\} \cup \{2\pi n - 1 + 2\pi : n \in \mathbb{Z}\}.$$

Correct answer. The solution set is $\emptyset$.

Explanation. For real u , $\cos u \in [-1, 1]$ and the principal arccosine has range $[0, \pi]$, so $\text{acos}(\cos u)$ is never negative. For example, a returned representative $u = 1$ gives $\text{acos}(\cos 1) - (-1) = 2$, not zero.

Likely root cause. The diagnosis identifies `sympy/solvers/solveset.py: _invert_real` applies $\text{acos}^{-1} = \cos$ to the target before checking that -1 lies in `acos`'s principal range. This is a new instance of a known failure mode.

Artifact (GitHub).

[bug-report/bug-031-false-real-solutions-for-acos-cos-u-1/](#)